

Yash Dave

yashdave@stanford.edu | github.com/yashdave003 | linkedin.com/in/yashdave003 | yashdave003.github.io

Education

Stanford University

June 2027

M.S. Computational and Mathematical Engineering, Data Science Track (3.8 GPA)

Coursework: AI Measurement Science, Numerical Linear Algebra, Convex Optimization, AI Reliability, ML from Human Pref.

University of California, Berkeley

May 2025

B.A. Applied Math (Economics cluster) and Computer Science (Honors); Public Policy Minor (3.94 GPA)

Coursework: ML*, Optimization*, Computer Vision*, Algorithms, Linear Algebra, Probability, Statistical Inference *graduate-level

Honors: Phi Beta Kappa, Outstanding GSI Award, Kevin K. Gong Memorial Scholarship , Upsilon Pi Epsilon, Dean's List

Publications

M. Hardy, S. T. Truong, A. Reuel, L. Zhang, J. M. Casabianca, **Y. Dave**, H. Lee, B. W. Domingue, S. Koyejo. *Noise Tectonics: Measuring the Stability of AI Benchmark Ecosystems*. International Conference on Machine Learning (ICML), accepted.

B. Marks*, **Y. Dave***, Z. Wang, H. Chung, R. Patwa, S. Cha, M. Murphy, A. Strang. *Do Generalized-Gamma Scale Mixtures of Normals Fit Large Image Datasets?*  arXiv:2512.17038, under review. *equal contribution

Research Experience

Stanford Trustworthy AI Research Lab

Stanford, CA

Research Assistant (under Prof. Sanmi Koyejo)

June 2025–present

- Building an expert-validated generative agent-based model of the AI evaluation ecosystem to study evaluator & policy design
- Exploring methods for sample-efficient prediction of agent reliability (cold-start IRT, TabPFN, etc.)

Strang Lab

Berkeley, CA

Research Assistant (under Prof. Alexander Strang)

January 2024–September 2025

- Developed testing pipeline to evaluate realism of a Bayesian mixture model  for flexibly modeling quasi-sparse image data
- Created a performant library  for generating CDFs of prior distributions at scale, finding best-fit parameters, and visualization

USAID Digital Workforce Development Project

Berkeley, CA

Consultant

Jan 2023–May 2023

- Led 12 group interviews with Cambodian professors to ascertain CS curriculum development needs and student experiences
- Synthesized findings to scope out potential syllabus revisions for 20 CS/DS courses to include best pedagogical practices

ReThink Media, Berkeley

Berkeley, CA

Data Science Intern

August 2022–December 2022

- Employed natural language inference and named entity recognition techniques on 400000+ quote dataset in Python
- Improved accuracy of speaker type classification models (Linear SVM, Random Forest, Logistic Regression) by 10%

Meet Your Future Project (Berkeley Economics and BRAC Uganda)

Berkeley, CA

Research Apprentice (under PhD Candidate Livia Alfonsi and Prof. Jeremy Magruder)

January 2022–October 2022

- Achieved 83% acc. in sentence-level topic classification of a 500+ transcript dataset using NLI-based zero-shot classification
- Performed sentiment analysis and emotion detection using VADER and NRC dictionary, creating visualizations using ggplot2

Teaching

Course Assistant, CS 229 Machine Learning , Stanford

Winter 2026

Course Assistant, CS 283 Governing Artificial Intelligence , Stanford

Autumn 2025

Teaching Assistant, Data 140 Probability  and Data 375 Pedagogy , UC Berkeley

Fall 2024–Spring 2025

Lead Content TA, Data 100 Principles of Data Science , UC Berkeley

Spring 2023–Spring 2024

- Co-authored public debugging guide  and contributing author of data science reference text 

Selected Projects

Direct Preference Optimization applied to Financial Advising

Dec 2025

- Fine-tuned a 4B-parameter model using DPO to automate SEC Marketing Rule compliance review, reducing reasoning tokens by 50% while maintaining expert-level rewrite quality.
- Built a privacy-preserving synthetic data pipeline generating 943 preference pairs from 40 expert annotations using a teacher-fixer-breaker framework to train compliant ad reviewers.

Beliefs about Govt. Intervention in India

May 2025

- Developed generalized linear regression model to study the causal relationship between local development and pro-intervention beliefs, controlling for factors like demographics, credit attribution, and government satisfaction